Asya Ünal

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Research interests: multimodal learning, generative models, agentic systems, AI evaluation

Anticipated graduation: 06/2027 · Applying to PhD programs, Fall 2027 · Available from Feb 2027

EDUCATION

Bilkent University, B.S. Computer Science *Sep 2023 – Jun 2027 (expected)*

GPA 3.65/4.00 · High Honor every semester · Merit Scholarship (60%) · All departmental coursework completed

City University of Hong Kong, Computer Science Exchange *Jan – May 2026*

Computer Vision, Artificial Intelligence, Data Science, Signals & Systems

Ankara University, Faculty of Medicine (M.D. program) *Sep 2020 – Sep 2023*

Three years of preclinical training (neuroanatomy, physiology, histology, biochemistry, pathology) before transferring to CS

RESEARCH EXPERIENCE

KUIS AI Center, Koç University *Sep 2026 – Present*

Undergraduate Researcher, advised by Prof. A. Murat Tekalp

- Studying semantic fidelity in diffusion super-resolution: whether text reconstructed by **DiffTSR** on **TextZoom** is actually supported by the low-resolution input, using OCR and vision-language models
- Fine-tuned **Qwen2.5-VL-3B** with **LoRA** as a reference-free judge of super-resolved candidates, keeping AI-curated supervision separate from human judgments
- Built reproducible train/eval pipelines with resumable checkpoints, source-grouped splits, strict and diagnostic scoring, and hash-verified artifacts

Department of Electrical and Electronics Engineering, Bilkent University *Sep 2026 – Present*

Undergraduate Researcher, advised by Prof. S. Serdar Kozat

- Working on agentic LLM orchestration for heterogeneous drone swarms (PX4, ArduPilot, INAV, DJI): mission decomposition, tool calls to optimization and CV/VLM modules, and replanning under GPS or communication loss
- Investigating GPS-denied autonomous flight that combines visual-inertial odometry with VLM/LLM planning, using small open models on lab drones and in simulation

Teknopar Industrial Automation *Jul – Aug 2024*

Machine Learning Research Intern

- Designed and evaluated a deep learning metal defect inspection pipeline across CNN (ResNet, Xception) and YOLO architectures, reaching **98%** test accuracy; first-author paper at IEEE FiCloud 2025
- Fine-tuned **DistilBERT** on 3K+ imbalanced industrial error logs, reaching **85%** accuracy

Institute of Neurological Sciences and Psychiatry, Hacettepe University *Sep 2022 – Feb 2023*

TÜBİTAK 2247-C STAR Intern Researcher

- Selected for TÜBİTAK's national STAR undergraduate research program; assisted with laboratory work on the role of Reduced Folate Transporter 1 in cerebral and retinal microcirculation

PUBLICATIONS

- **A. Ünal** et al., "AI-Driven Collaborative Assembly Line Inspection: System Integration and Deployment Challenges," *MobiWIS 2026*, Granada, Spain. arXiv:2609.33522
- **A. Ünal** et al., "Surface and Edge Defect Inspections on Metals Using Deep Learning and Optical Camera Data," *IEEE FiCloud 2025*, Istanbul, Turkey.
- F. Kaya, **A. Ünal** et al., "Real-Time Anomaly Detection in Industry 4.0 Using Asset Administration Shell," *Springer LNCS 16066*, MobiWIS 2025, pp. 114–128.

ENGINEERING EXPERIENCE

AlgoritiX (CodeAPeel), with Prof. A. Yavuz Oruç, University of Maryland *Aug 2026 – Present*
Programmer, part time

- Engineered a FastAPI AI assistant with an execute, observe, repair loop: generated assembly runs in a bounded sandbox and is verified before any editor change is proposed
- Lead development of Java desktop tools and a React/TypeScript workspace for MIPS, RISC-V, VESP and Ark-U with execution, memory inspection and test-backed challenges

Mobile Action *Jun – Jul 2025*
Software Engineering Intern

- Initiated and built a Model Context Protocol server exposing 27 App Store analytics endpoints as LLM agent tools; shipped SearchAds features in Vue 3, Spring Boot and RabbitMQ

Eluvium, London *May – Jun 2025*
AI Engineer Intern

- Re-architected a monolithic state machine into a modular LangGraph workflow inside a multi-service production codebase, shipping through code review with senior engineers
- Built LLM agent components on Gemini, including a LangChain classifier that routes manufacturing alerts to the right handler and reduced manual triage

SOCAR, Istanbul *Aug – Sep 2024*
Software Developer Intern

- Built a service management platform (Spring Boot, React, PostgreSQL) with real-time currency pricing via a SOAP API

SELECTED PROJECTS

- **Miper** [miper.ai], founder and sole engineer: architecting an agentic AI platform where specialist agents share one personal knowledge graph (journals, voice, photos, email) to plan, track and complete a person's work. Its ingestion layer admits only LLM-extracted facts anchored to exact source spans and reconciles contradicting updates over time
- **Qardex** [qardex.app], founder: Flutter app that turns notes into adaptive flashcards and live multiplayer study races

HONORS & AWARDS

- Bilkent University High Honor Student, every semester 2023 – Present
- Bilkent University Merit Scholarship, increased from 40% to 60% 2024 – Present
- TÜBİTAK 2247-C STAR Intern Researcher Scholarship 2022
- Ranked 2,670th of 2.4 million (top 0.11%) in Turkey's national university entrance exam 2020
- TED Ankara College Comprehensive Scholarship 2016 – 2020

LEADERSHIP

ACM Bilkent Student Chapter *Sep 2024 – Present*
President (2024–25), Head of Audit Committee (2025–26)

- Led a 400+ member chapter; grew event participation from 207 to 911 in one year and organized the Entrepreneurship Summit and the Amazon University Engagement Program

SKILLS

ML & AI: PyTorch, Hugging Face, LoRA, diffusion models, VLMs, computer vision, LLM agents, LangGraph, MCP, RAG
Programming & Systems: Python, C++, C, Java, TypeScript, FastAPI, React, PostgreSQL/pgvector, Docker
Languages: Turkish (native), English (C2, IELTS 8.5/9), German (B1) **Interests:** Sailing (since 2011)